

Section: Our Dynamic Universe

Topic: Forces, Energy and Power

Question 2 – Pushing a Wheelie Bin Up a Slope

(a) Calculate the component of weight acting down the slope

✓ 2 marks

Given:

- Mass of bin: $m = 24.0 \text{ kg}$
- Gravitational field strength: $g = 9.8 \text{ m/s}^2$
- Slope angle: $\theta = 10^\circ$

Use:

$$F = mg \sin \theta = 24.0 \times 9.8 \times \sin(10^\circ)$$

$$F = 235.2 \times 0.1736 = 40.8 \text{ N}$$

✓ Answer: 40.8 N

(b) Explain why the frictional force is not included in this calculation

✓ 1 mark

✓ Answer:

The question only asks for the **component of weight**, which is **due to gravity**.

Friction is a **separate force** that opposes motion but is not part of the gravitational component.

Revision Tips

- Weight down a slope:

$$F = mg \sin \theta$$

- $\sin \theta$ increases with steeper slopes
 - Friction acts **parallel** to the surface and **opposes motion**, not included in gravitational component unless specified
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